CONCEPT 56.1

Human activities threaten earth's biodiversity

Throughout the biosphere, human activities are altering natural disturbances, trophic structures, energy flow, and chemical cycling—ecosystem processes on which we and all other species depend. We have physically altered nearly half of Earth's land surface, and we use over half of all accessible surface fresh water. In the oceans, stocks of most major fisheries are shrinking because of overharvesting. By some estimates, we may be pushing more species toward extinction than did the asteroid that triggered the Cretaceous mass extinction 66 million years ago (see Figure 25.18).

In this chapter, we'll examine changes happening across Earth, focusing on **conservation biology**, a discipline that integrates ecology, physiology, molecular biology, genetics, and evolutionary biology to conserve the diversity of life on Earth. Efforts to sustain ecosystem processes and stem the extinction of species also connect the life sciences with the social sciences, economics, and humanities.

Extinction is a natural phenomenon that has been occurring since life first evolved; it is the high *rate* of extinction that is of concern. More than 1,000 species have become extinct in the last 400 years, a rate that is 100 to 1,000 times the "background," or typical, extinction rate seen in the fossil record (see Concept 25.4). This comparison indicates that the extinction rate today is high and that human activities threaten Earth's biological diversity at all levels.

Three Levels of Biodiversity

Biodiversity—short for biological diversity—can be considered at three main levels: genetic diversity, species diversity, and ecosystem diversity **(Figure 56.2)**.

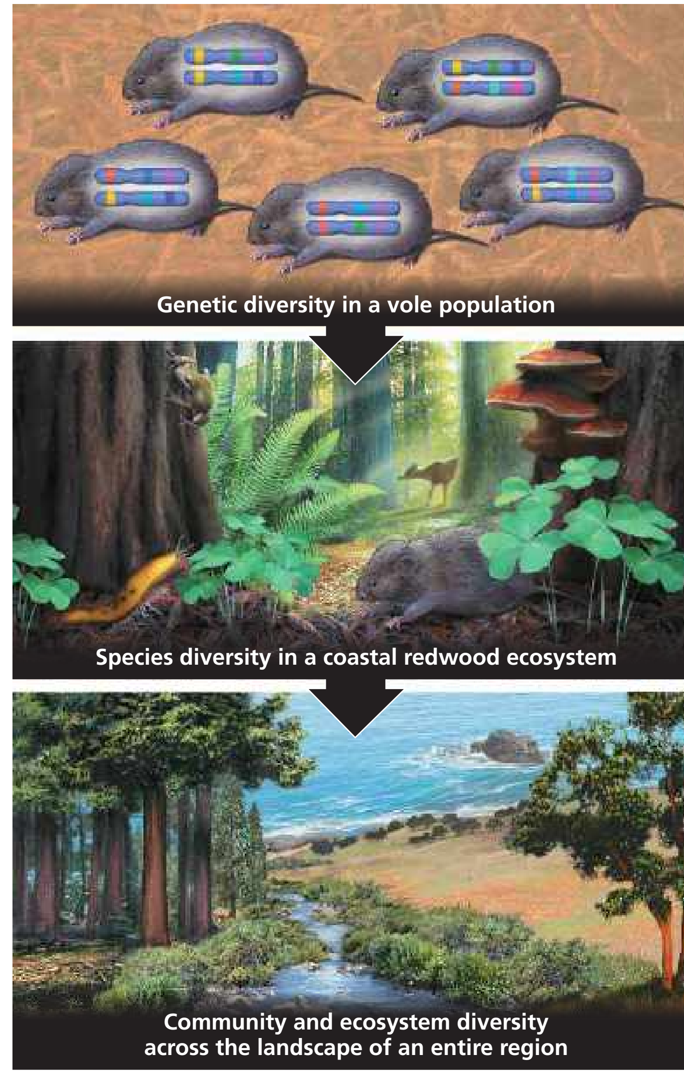
Genetic Diversity

Genetic diversity comprises not only the individual genetic variation *within* a population, but also genetic differences *between* populations—differences that often are associated with adaptations to local conditions. If one population becomes extinct, then a species may have lost some of the genetic diversity that makes microevolution possible. This erosion of genetic diversity in turn reduces the adaptive potential of the species.

Species Diversity

Public awareness of the biodiversity crisis centers on species diversity—the number of species in an ecosystem or across the biosphere. To date, scientists have described and formally named about 1.8 million species of organisms. In addition to these named species, many others remain to be discovered:

▼ **Figure 56.2 Three levels of biodiversity.** The oversized chromosomes in the top diagram symbolize the genetic variation within the population.



Estimates for the number of species that currently exist range from 5 million to 100 million.

Of particular concern are species that are endangered or threatened. An **endangered species** is one that is in danger of extinction throughout all or much of its range **(Figure 56.3)**, while a **threatened species** is considered likely to become endangered in the near future. The following are just a few statistics that illustrate the problem of species loss:

- According to the International Union for Conservation of Nature and Natural Resources (IUCN), 13% of the 10,000 known species of birds and 22% of the 5,500 known species of mammals are threatened.
- A survey by the Center for Plant Conservation showed that of the nearly 20,000 known plant species in the United States, 200 have become extinct since such records have been kept, and 730 are endangered or threatened.